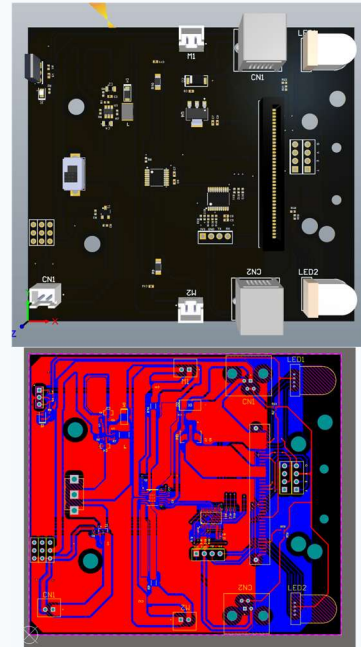


Project 5: Custom Embedded Controller Board

A compact embedded control hardware solution engineered for localized communication and peripheral tracking.

- Form Factor & Layout: Highly integrated black soldermask PCB with sleek 3D components optimization.
- Interfaces: Onboard CN1/CN2 RJ45/Ethernet-style communication ports and dual LED indicators.
- Processing: Central microcontroller hub mapped with precise crystal oscillators and filtering caps.

[ [PASTE EMBEDDED CONTROLLER 3D VIEW HERE](#)]
(Screenshot 2026-06-16 124415_2.png)



Project 6: Dual-Layer Control Routing Layout

The structured 2D copper layout design matching the compact embedded controller architecture.

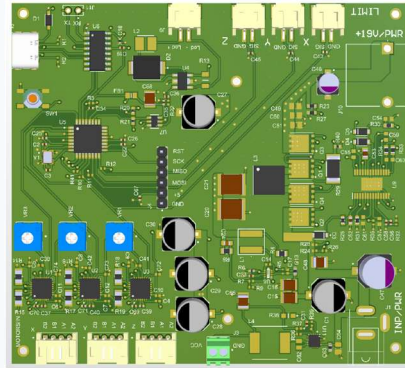
- Power Routing: Solid red copper pours on top layer for enhanced power paths and EMI masking.
- Signal Integrity: Isolated trace paths (blue traces) on bottom layer to bypass critical cross-talk.
- Manufacturing Clearances: Precision clearance rules for high-yield SMD component placement.

[ [PASTE 2D ROUTING LAYOUT HERE](#)]
(Screenshot 2026-06-16 124425_2.png)

Project 7: Multi-Axis CNC & Motor Driver Shield

An advanced industrial-grade controller board designed for multi-axis movement tracking and CNC operations.

- Motion Control: Integrated 3-Axis (X, Y, Z Axis) motor terminal logic with onboard limit sensor headers.
- Power Conditioning: Robust buck-regulation circuitry supporting stable +19V/PWR power inputs.
- Tuning & Isolation: Embedded precise pot-tuned motor drivers with heavy bulk capacitors for peak current damping.



**PASTE CNC MOTOR DRIVER
SCREENSHOT HERE]**
(Screenshot 2026-06-16 124830.png)